IBM Developer
SKILLS NETWORK

Winning Space Race with Data Science

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Outline

Executive summary and research question
Data collection and preparation
Exploratory analysis with charts and SQL
Interactive maps and dashboard results
Classification models and evaluation
Conclusions, limitations and reproducibility

Executive Summary

90 launches: 60 positive landing labels (66.7%).
All four classifiers reach 15/18 correct test predictions (83.3%).
Decision tree leads cross-validation at 87.68%.
Site, orbit and launch history show descriptive differences.
A small test set and preprocessing leakage limit generalization.

Introduction

Research question: can historical launch attributes predict the first-stage landing label?

Reusability provides the business motivation. This project predicts landing outcomes rather than launch prices.

Scope: historical IBM teaching datasets. Target: binary landing Class. Mission success is a separate variable.

Section 1

Methodology



Methodology

Collect historical launch data through a REST API and HTML tables.
Clean records, define landing labels and encode categorical features.
Use SQL and exploratory charts to investigate patterns.
Use Folium and Dash to explore locations and filtered outcomes.
Tune four classifiers with 10-fold cross-validation, then evaluate 18 held-out launches.

Data Collection

Source	Role	Analysis population
SpaceX API teaching snapshot	Launch details and resource lookups	90 modeled Falcon 9 launches
Wikipedia revision, 9 June 2021	Historical launch table extraction	Separate scraped export
Spacex.csv	SQL exploration	101 historical records
Geo and Dash teaching CSVs	Interactive visual analytics	56 launches

The datasets have different scopes. Counts and success rates must retain their denominators.

Data Collection: SpaceX API

Step	Implementation
Request and parse	GET snapshot, response.json(), pd.json_normalize()
Enrich identifiers	Resolve rocket, launchpad, payload and core IDs
Restrict population	One payload and one core, date through 13 Nov 2020, Falcon 9
Prepare export	Fill missing payload masses with the mean, retain missing LandingPad
Output	dataset_part_1.csv, followed by 90-row modeling snapshot

API notebook completed. Report statistics use the supplied downstream teaching snapshot.

Data Collection: Web Scraping

Step	Implementation
Retrieve HTML	Historical Wikipedia page rather than the evolving live table
Locate launch records	BeautifulSoup, table headers and launch-number rows
Clean cell text	Remove reference annotations, normalize text and mass strings
Build dataset	Flight number, date/time, booster, site, payload, orbit, outcome
Output	spacex_web_scraped.csv

The scraping notebook is completed. Its export is separate from the modeling sample.

Data Wrangling

Operation	Result / rationale
Inspect missing values	LandingPad: 26/90 missing in the API teaching dataset
Payload mass imputation	Replace missing values with the observed mean
Create target	60 positive and 30 negative labels
One-hot encoding	Orbit, LaunchSite, LandingPad and Serial
Feature matrix	90 rows × 80 float64 predictors, target stored separately

Lab convention counts True Ocean as positive. Successful recovery for reuse is a stricter business target.

EDA with Data Visualization

Scatterplots: launch number and payload against site and orbit, colored by Class.
Bar chart: mean Class by orbit, with sample counts.
Line chart: mean Class by launch year.

These views describe associations. Site, mission type and technology also change over time.

EDA with SQL

DISTINCT and LIKE identify launch sites and matching records.
SUM and AVG quantify customer payload totals and booster payload means.
MIN identifies the first successful ground landing.
GROUP BY counts mission and landing outcomes.
Subqueries identify all boosters carrying the maximum payload.
Date filters isolate 2015 failures and rank outcomes over a fixed period.

Interactive Map Methodology

Folium circles and labels locate each launch site.
MarkerCluster groups overlapping launch records. Green means class 1, red means class 0.
MousePosition displays coordinates for exploration.
Haversine distances and PolyLine connect sites to selected study points.

Distance results are straight-line reference measurements, not validated shortest distances to infrastructure.

Plotly Dash Methodology

Searchable dropdown: ALL or one launch site.
ALL: pie chart of successful launch counts by site.
Single site: pie chart of class 0 versus class 1.
Payload slider: 0 to 10,000 kg, in 1,000 kg steps.
Scatterplot: payload versus Class, colored by booster version.
The pie responds to the site dropdown. The scatter responds to both inputs.

Predictive Analysis Methodology

Stage	Configuration
Input	90 records, 83 encoded features, binary Class
Preprocessing	StandardScaler fitted before split, as required by lab
Split	train_test_split(test_size=0.2, random_state=2)
Training	72 records, GridSearchCV with cv=10 and accuracy scoring
Models	Logistic regression, SVM, decision tree, KNN
Evaluation	18 test records, accuracy and confusion matrices

A production evaluation should fit preprocessing inside each cross-validation training fold.

Results

- EDA: outcome patterns across orbit, site, payload and time.
- SQL: query outputs with explicit filters and population scope.
- Interactive analytics: geographic layers and dashboard selections.
- Classification: test accuracy, validation performance and error analysis.

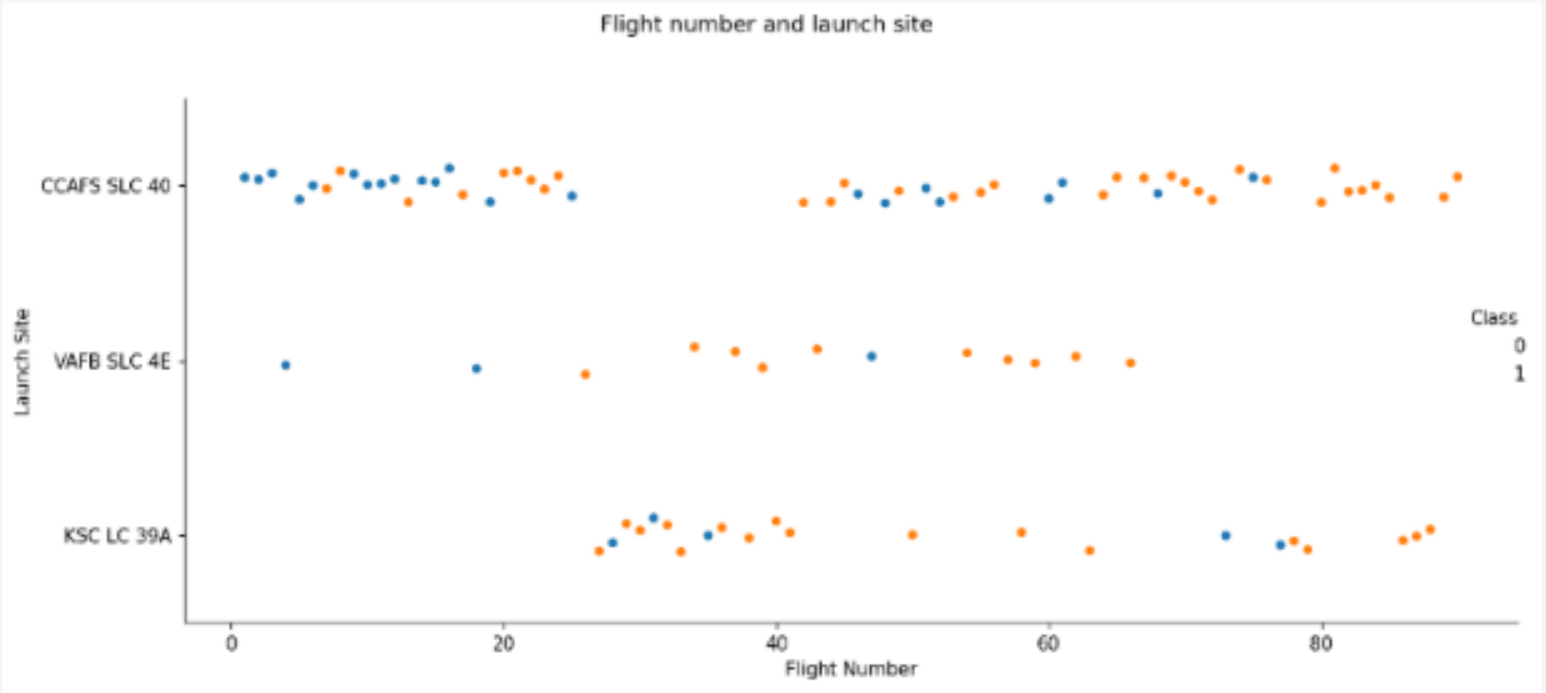
Section 2

Insights drawn from EDA

Flight Number vs. Launch Site

Later launches contain more successful outcomes.

CCAFS SLC 40: 55 launches
KSC LC 39A: 22 launches
VAFB SLC 4E: 13 launches



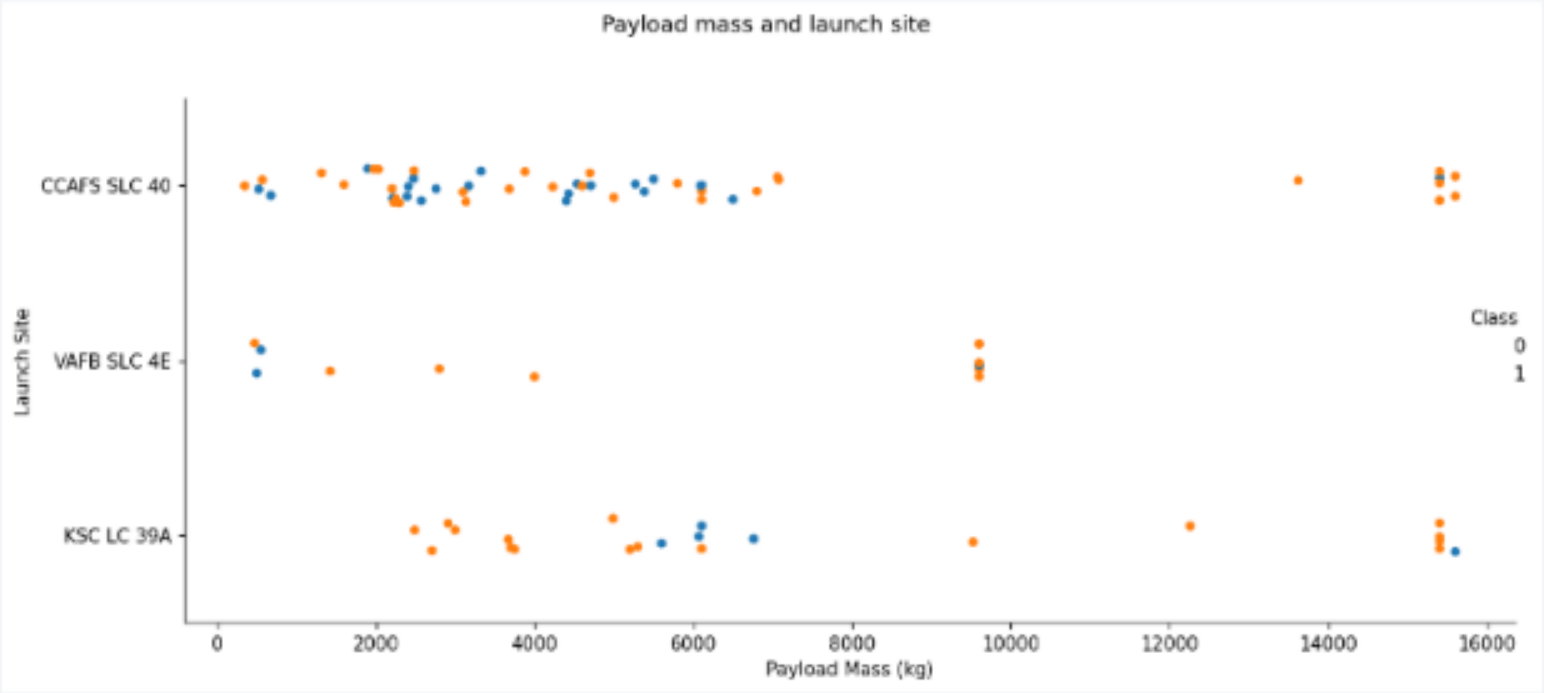
90-row modeling dataset. Class 0 and Class 1 appear in the original notebook plot legend.

Payload vs. Launch Site

The observed payload distribution differs by site.

VAFB has no payload above 10,000 kg in this sample.

These differences may reflect mission assignment.

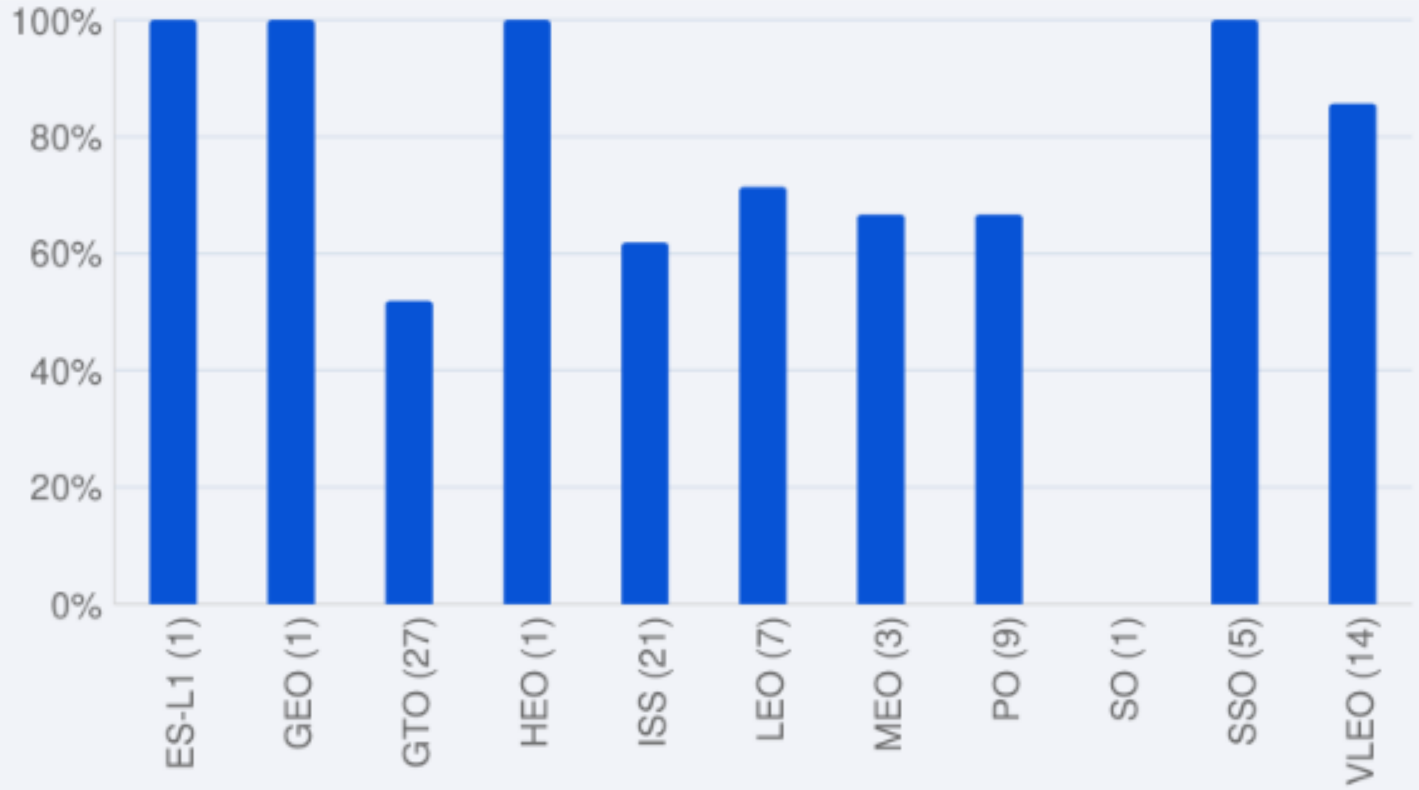


Payload mass is measured in kilograms. Descriptive association, not a causal site effect.

Success Rate vs. Orbit Type

VLEO: 12/14 (85.7%).
GTO: 14/27 (51.9%).
SSO: 5/5 (100%).

Several perfect rates come from only one launch.



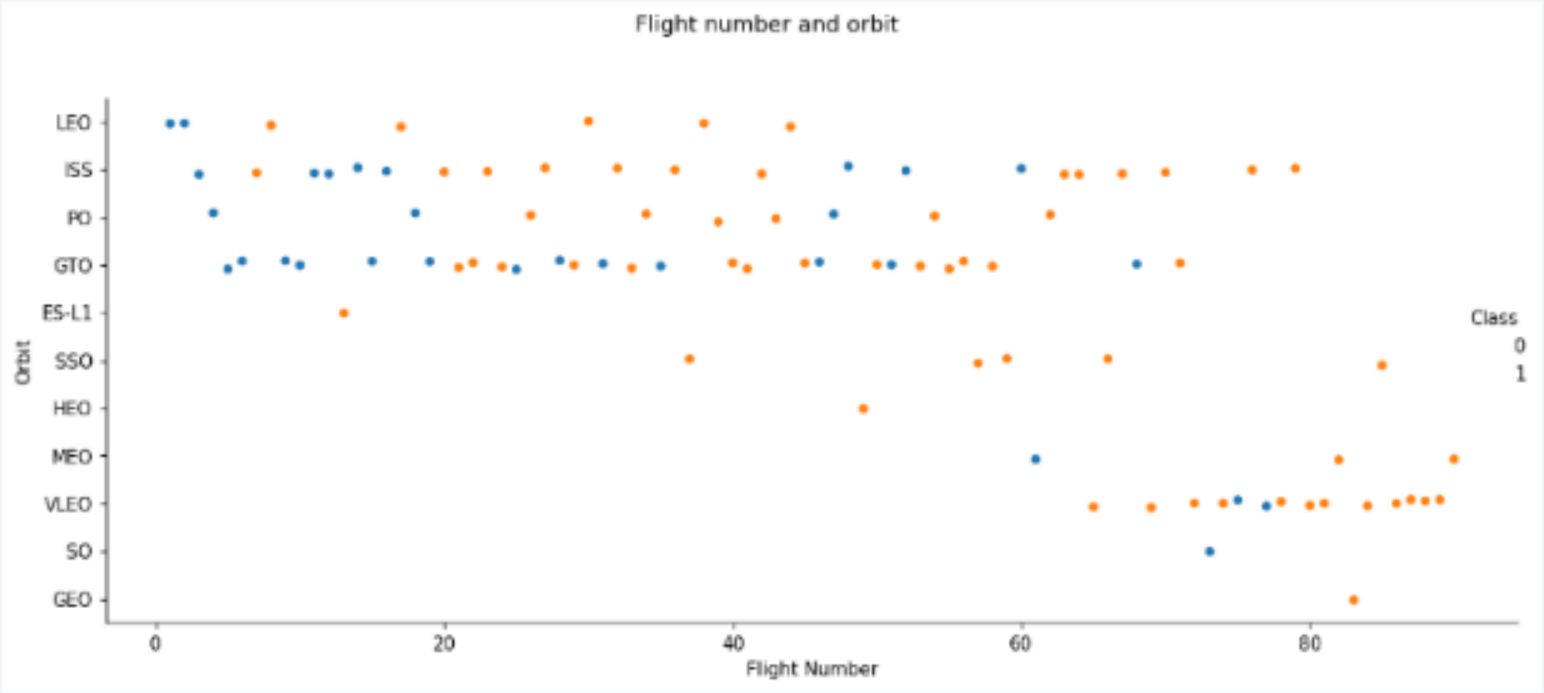
Labels show orbit and n. GTO is a transfer orbit, retained here for a complete mission-type comparison.

Flight Number vs. Orbit Type

LEO launches show more positive labels later in the sequence.

GTO includes both outcomes throughout the observed period.

Flight number also proxies technological progress.



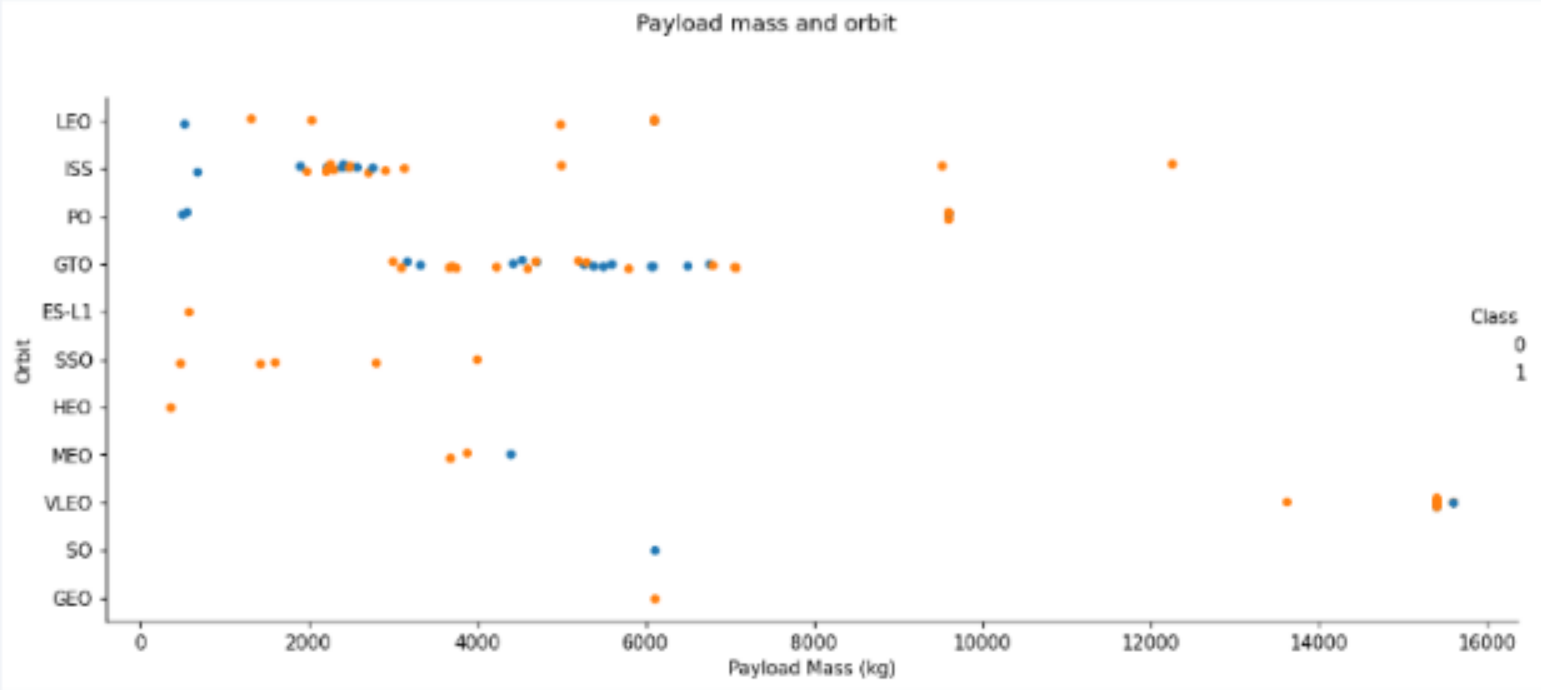
Launch sequence and calendar time overlap. A random split may overstate future performance.

Payload vs. Orbit Type

Payload and orbit jointly describe the mission profile.

Successes and failures overlap in GTO.

A single payload threshold cannot separate all outcomes.



90 launches. Use the full feature context rather than treating payload as an independent causal driver.

Launch Success Yearly Trend

2014 and 2015: 33.3%.
2017: 83.3%.
2018: 61.1%.
2020: 84.2%.

Improvement is uneven.



Annual counts range from 1 to 19. No observation for 2011 in the modeling dataset.

All Launch Site Names

Distinct Launch_Site values
CCAFS LC-40
VAFB SLC-4E
KSC LC-39A
CCAFS SLC-40

Four labels in the SQL dataset.

The 90-row dataset groups launches into three site labels.

```
SELECT DISTINCT Launch_Site FROM SPACEXTABLE;
```

Launch Sites Beginning with 'CCA'

Date	Booster	Site	Payload kg
2010-06-04	F9 v1.0 B0003	CCAFS LC-40	0
2010-12-08	F9 v1.0 B0004	CCAFS LC-40	0
2012-05-22	F9 v1.0 B0005	CCAFS LC-40	525
2012-10-08	F9 v1.0 B0006	CCAFS LC-40	500
2013-03-01	F9 v1.0 B0007	CCAFS LC-40	677

WHERE Launch_Site LIKE 'CCA%' ORDER BY Date LIMIT 5. Dates ensure reproducible record selection.

Total Payload Mass: NASA CRS

45,596 kg

Total payload for Customer = 'NASA (CRS)'.
This is a customer-filtered sum in the SQL teaching dataset.

SELECT SUM(PAYLOAD_MASS__KG_) ... WHERE TRIM(Customer) = 'NASA (CRS)';

Average Payload Mass: F9 v1.1

2,928.4 kg

Average payload for the exact Booster_Version value 'F9 v1.1'.
The query does not include every string beginning with that version label.

Exact match filter retained from the lab. A LIKE 'F9 v1.1%' query defines a different population.

First Successful Ground Landing

22 December 2015

MIN(Date) among records whose Landing_Outcome is Success (ground pad).

ISO dates allow chronological MIN(Date) in SQLite. Landing outcome differs from mission outcome.

Successful Drone Landings: 4,000–6,000 kg

Matching booster versions
F9 FT B1022
F9 FT B1026
F9 FT B1021.2
F9 FT B1031.2

Strict conditions: payload > 4,000 AND payload < 6,000. Landing_Outcome = Success (drone ship).

Mission Outcome Counts

Mission result	Count
Success (normalized)	100
Failure (in flight)	1
Total	101

Normalize whitespace and group Success-prefixed variants.

Mission success does not imply first-stage landing success.

Normalized CASE expression groups Success (payload status unclear) with Success.

Boosters Carrying the Maximum Payload

15,600 kg

Booster version	Booster version	Booster version
F9 B5 B1048.4	F9 B5 B1049.4	F9 B5 B1051.3
F9 B5 B1056.4	F9 B5 B1048.5	F9 B5 B1051.4
F9 B5 B1049.5	F9 B5 B1060.2	F9 B5 B1058.3
F9 B5 B1051.6	F9 B5 B1060.3	F9 B5 B1049.7

Subquery: WHERE PAYLOAD_MASS__KG_ = (SELECT MAX(PAYLOAD_MASS__KG_) FROM SPACEXTABLE).

Drone Ship Landing Failures in 2015

Date	Booster version	Launch site
2015-01-10	F9 v1.1 B1012	CCAFS LC-40
2015-04-14	F9 v1.1 B1015	CCAFS LC-40

Two matching records in this filtered SQL sample.

WHERE Date LIKE '2015%' AND Landing_Outcome = 'Failure (drone ship)'.

Landing Outcomes: June 2010 to March 2017

Outcome	Count
No attempt	10
Success (drone ship)	5
Failure (drone ship)	5
Success (ground pad)	3
Controlled (ocean)	3
Uncontrolled (ocean)	2
Failure (parachute)	2
Precluded (drone ship)	1

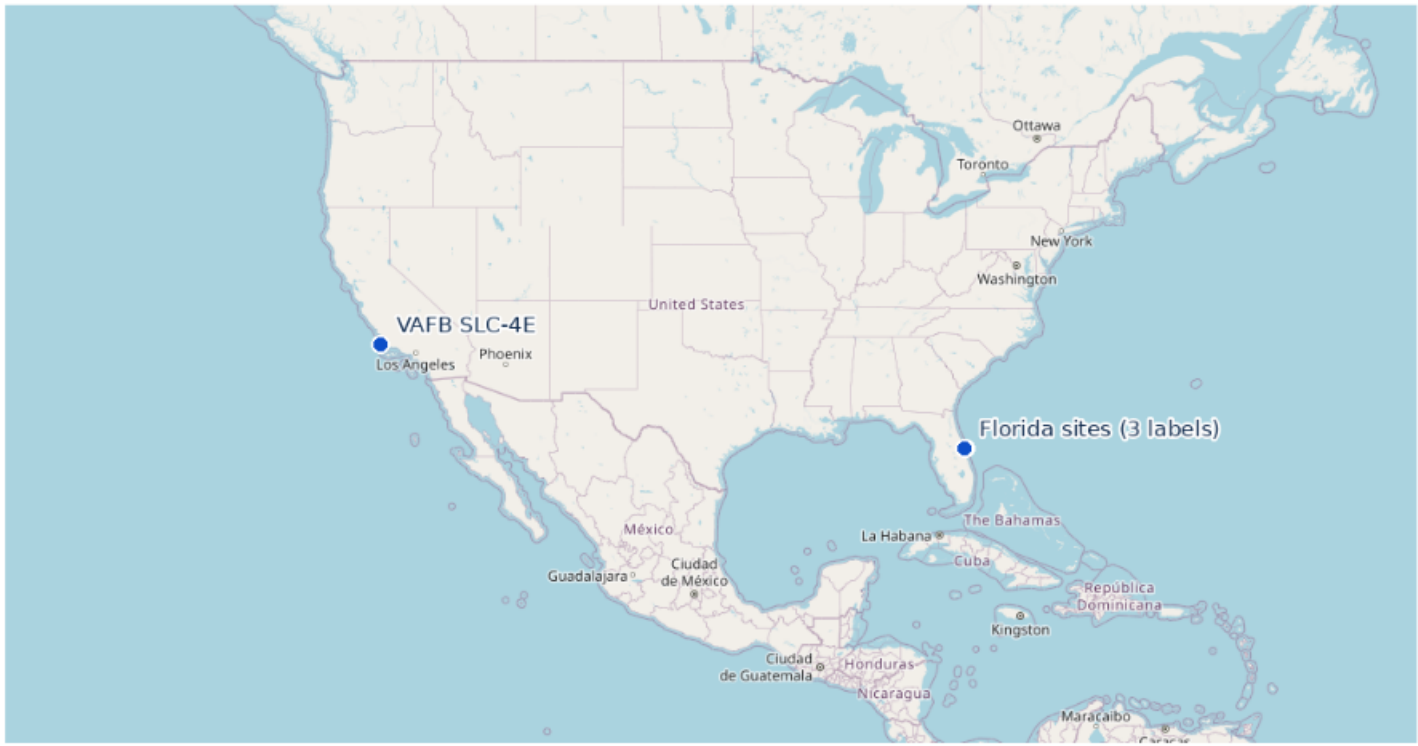
Inclusive period: 2010-06-04 to 2017-03-20. 31 records. Ties are ordered alphabetically.

Section 3

Launch Sites Proximities Analysis

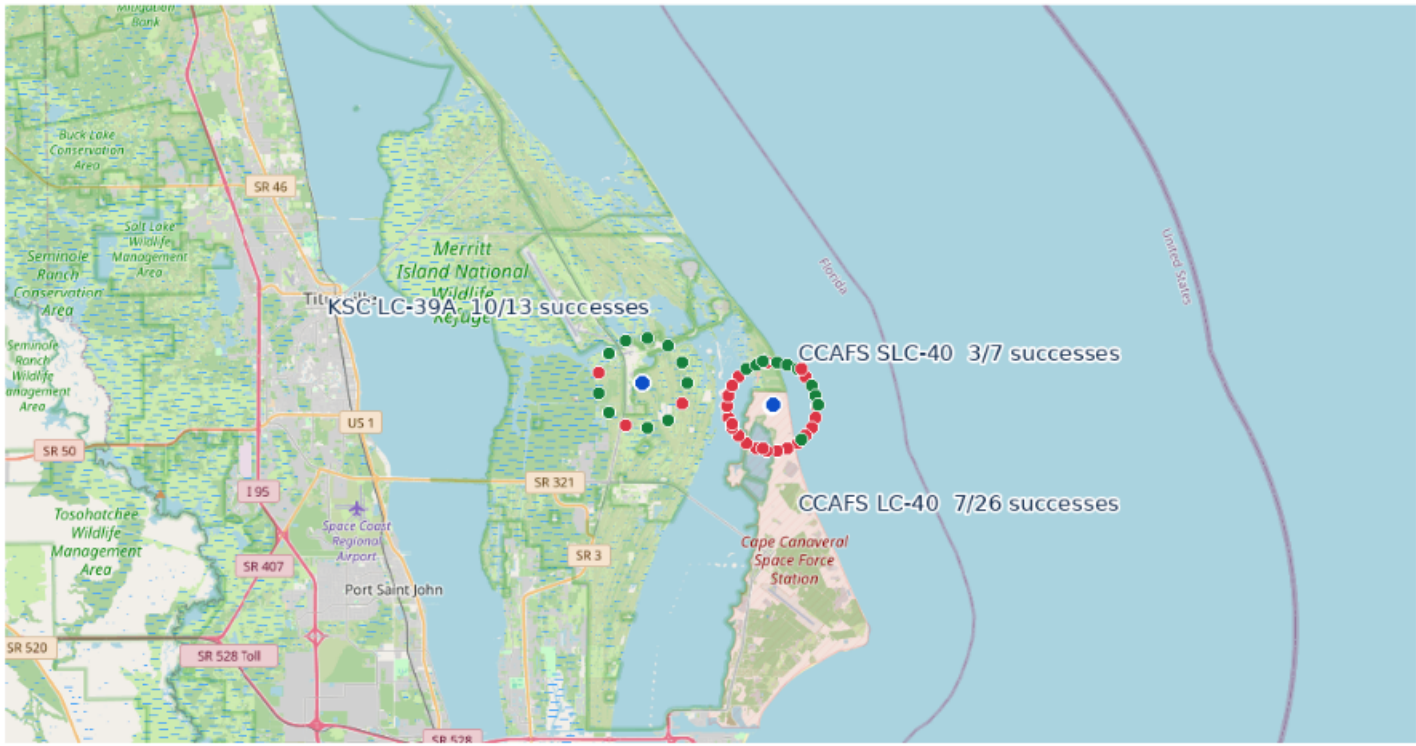


Folium Results: Launch Site Locations



Static export of Folium coordinates. Four site labels, on the Florida and California coasts.

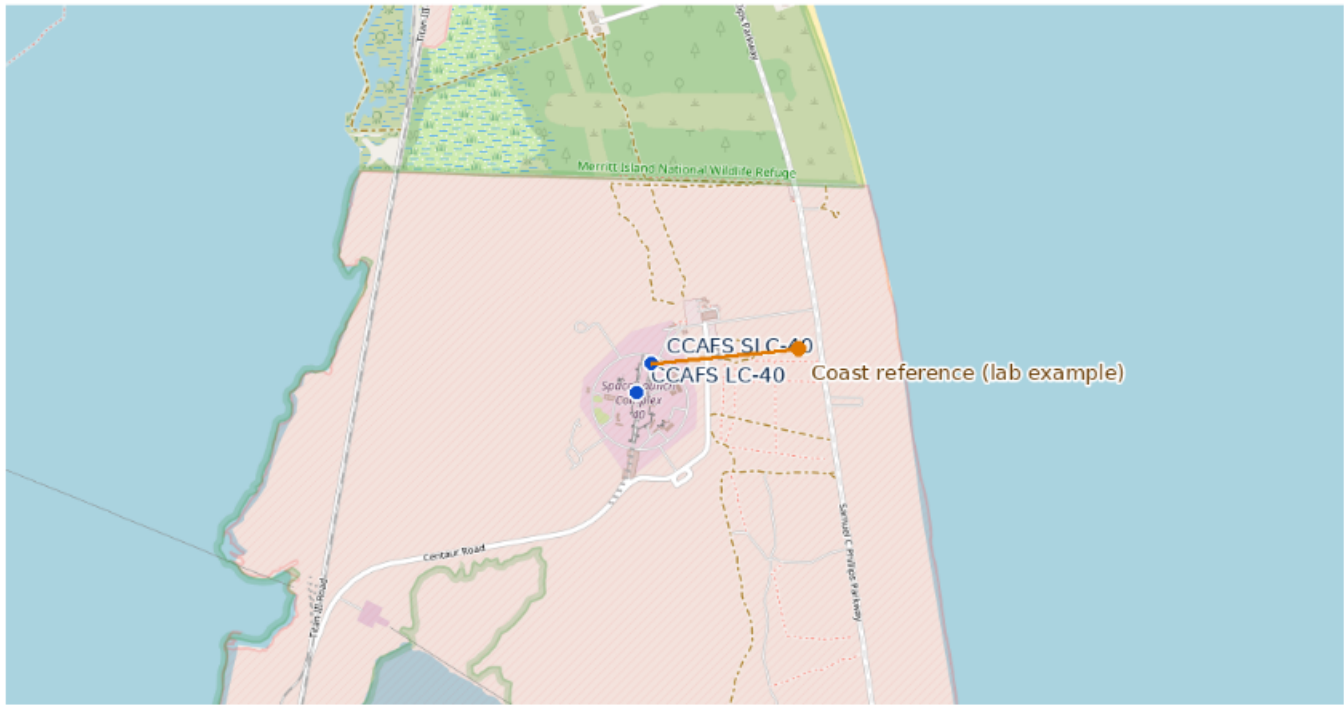
Folium Results: Co-located Launch Outcomes



Static export of Folium layer coordinates. Map: © OpenStreetMap contributors

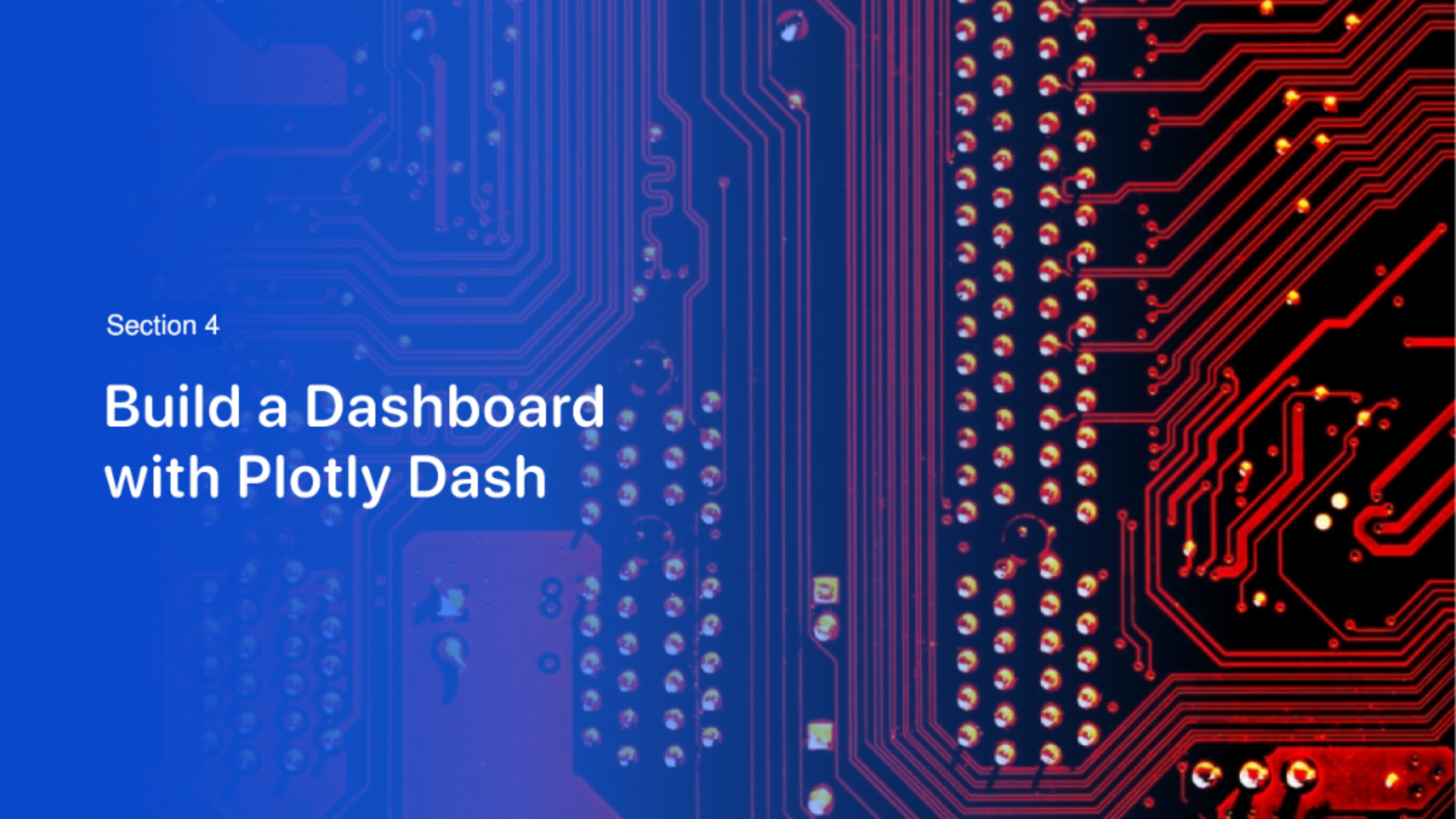
Static export, 56-launch sample. Green = class 1, red = class 0. Markers spread for visibility.

Folium Results: Coastline Reference Distance

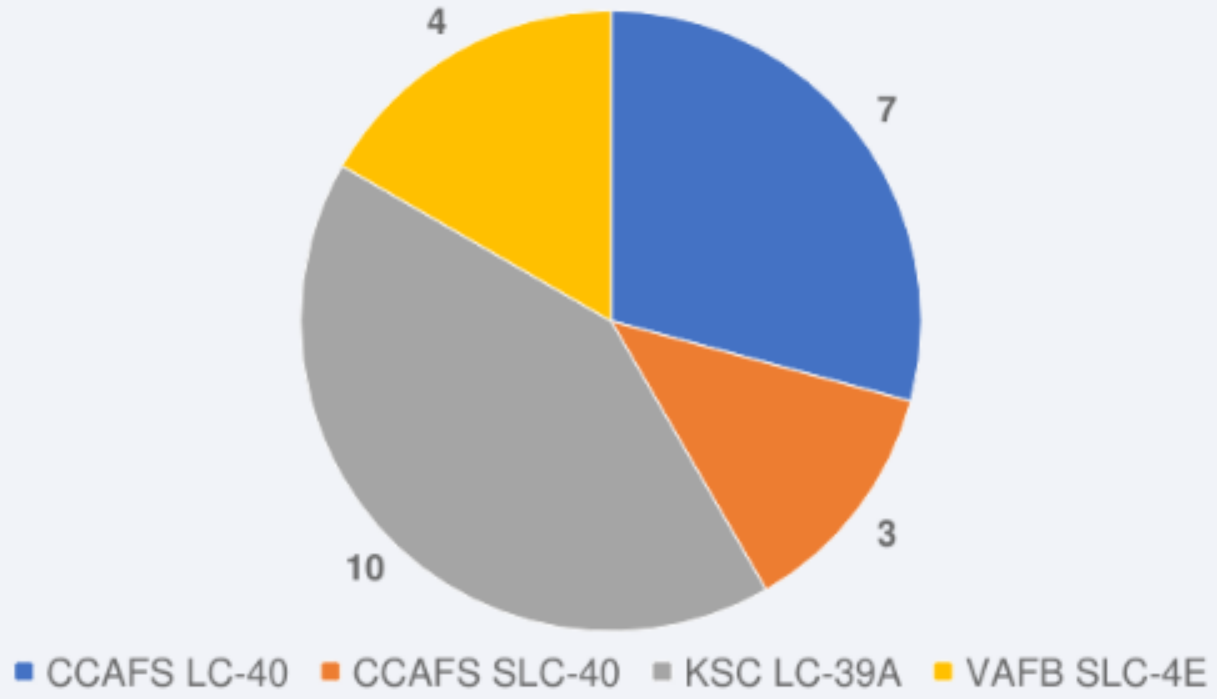


CCAFS SLC-40 to the lab coastline reference: about 0.51 km, at a point selected for the exercise.

Approximate study coordinate. This is not a validated minimum distance to the coast or infrastructure.


Section 4
Build a Dashboard with Plotly Dash

Dash Results: Successful Launches by Site



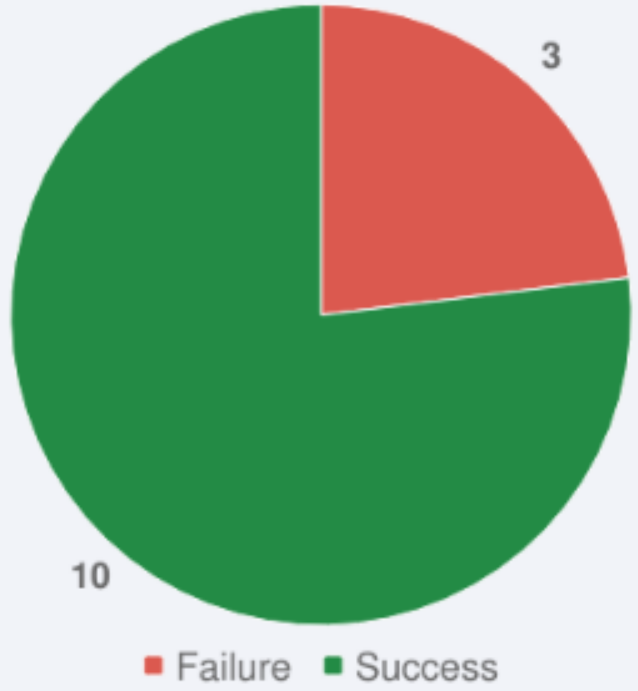
Selection: ALL sites

KSC LC-39A contributes 10 of 24 positive outcomes (41.7%).

Largest success count in this sample.

Static chart of the dashboard callback result. Pie values count successes, not all launches.

Dash Results: KSC Has the Highest Success Rate



Selection: KSC LC-39A

10 successes / 13 launches
76.9% success rate

Sample: 56 total launches.

Static chart of the selected-site dashboard callback result.

Dash Results: Payload Filters and Booster Versions

Selection / comparison	Successes / n	Rate
Payload >3,000 to 4,000 kg	8 / 11	72.7%
Payload >6,000 to 7,000 kg	0 / 4	0.0%
Booster B5	1 / 1	100.0%
Booster FT	16 / 24	66.7%

Rates depend on the selected bins and small denominators. B5 has only one observation.

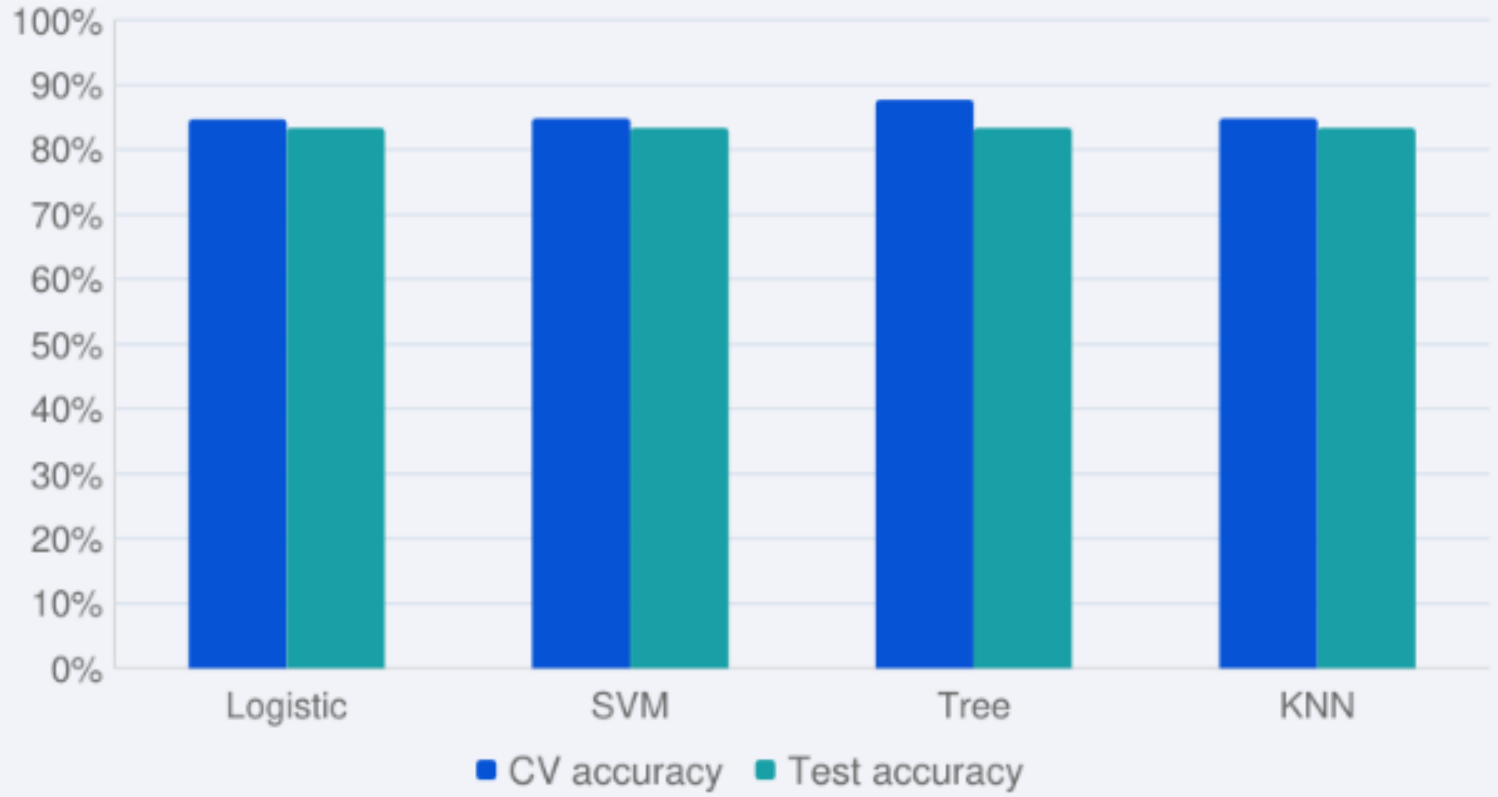
Dash dataset, 56 launches. Fixed 1,000 kg bins, lower boundary excluded and upper included.

Section 5

Predictive Analysis (Classification)



Classification Accuracy



All models: 83.3% test accuracy.

Decision tree: highest CV score, 87.68%.

There is no unique test winner.

10-fold validation on 72 training records. Test contains 18 launches. CV and test scores serve different roles.

Confusion Matrix: Decision Tree

Actual / predicted	Did not land	Landed
Did not land	3	3
Landed	0	12

True positives: 12
True negatives: 3
False positives: 3
False negatives: 0

Positive precision: 80%
Positive recall: 100%

Selected by CV score. False positives create overconfidence in landing success. Results use the lab label.

Conclusions

The historical dataset supports a complete collection-to-model workflow.
EDA shows patterns across site, orbit and time, with confounding and small groups.
All four classifiers reach 83.3% test accuracy. The decision tree leads validation.
The landing label and intended business use need closer alignment.
Next evaluation: chronological split, fold-specific preprocessing and as-of-launch features.

Appendix: Dataset Scope and Reproducibility

Artifact	Scope / output
API and wrangling	90-launch teaching modeling snapshot
EDA / feature engineering	90 × 80 feature matrix, 60 positive labels
SQL exploration	101 records, mission and landing outcomes
Folium / Dash	56 launches, four site labels
Machine learning	72 train / 18 test, random_state = 2

Different source files can produce different counts without contradiction.

Acknowledgements

IBM Skills Network supplied the capstone labs, teaching datasets and presentation template.

Analysis, completed code and interpretation:
Emmanuel BONKOUNGOU

Technical appendices follow.

Appendix: Hyperparameter Search

Logistic regression: C = 0.01, 0.1, 1. L2 penalty, lbfgs solver.
SVM: linear, rbf, poly, sigmoid. Five C and gamma values from 10^-3 to 10^3. Best: sigmoid, C = 1, gamma = 0.03162.
Tree: gini / entropy, best / random splits, depths 2–18, sqrt features, leaf sizes 1/2/4, split sizes 2/5/10. Best: gini, depth 8, leaf 2, split 10.
KNN: k = 1–10, four algorithms, p = 1 or 2. Best: k = 10, p = 1, auto.

Innovative Insights and Evaluation Risks

15/18 correct yields an approximate 95% Wilson accuracy interval of 60.8%–94.2%. One case changes accuracy by 5.56 points.
The majority baseline is 12/18 (66.7%). Models improve by three correct predictions on this small test.
83 predictors versus 72 training records increases overfitting risk. Serial indicators may encode identity and time.
True Ocean counts as a positive lab label, while reusable recovery needs a stricter target.
Current API reuse counts need an as-of-launch audit.

Appendix: Source Files and References

IBM Skills Network course template and completed capstone notebooks.
Modeling: datasets/dataset_part_2.csv and dataset_part_3.csv.
Interactive analytics: datasets/spacex_launch_geo.csv and spacex_launch_dash.csv.
SQL: labs/module_2/data/Spacex.csv.
Base URL:
https://cf-courses-data.s3.us.cloud-object-storage.appdomain.cloud/IBM-DS0321EN-SkillsNetwork/
Maps: OpenStreetMap contributors. All claims refer to historical teaching snapshots.

GitHub Repository and Project Files

GITHUB REPOSITORY LINK: TO BE ADDED

Upload all completed notebooks and Python files to your SpaceX repository.
Add the exact public repository URL here before submission.

Include the API, scraping, wrangling, EDA, SQL, Folium and ML notebooks, plus spacex_dash_app.py, data files and requirements.txt.

Repository URL left pending at the author's request. This grading item is not yet satisfied.